

Watchy v3 — Test Point Identification

Multimeter procedure for the GDEY0154D67-FL04 frontlight mod. Written for a WH5000A autoranging meter; any DMM with continuity, DC volts and resistance will do.

1. What you end up with

Three pads on the Watchy board, positively identified and marked: **TP4 (3V3)**, **TP5 (GND)** and **TP2 (PWM, GPIO44)**. Those are the only connections the frontlight adapter needs.

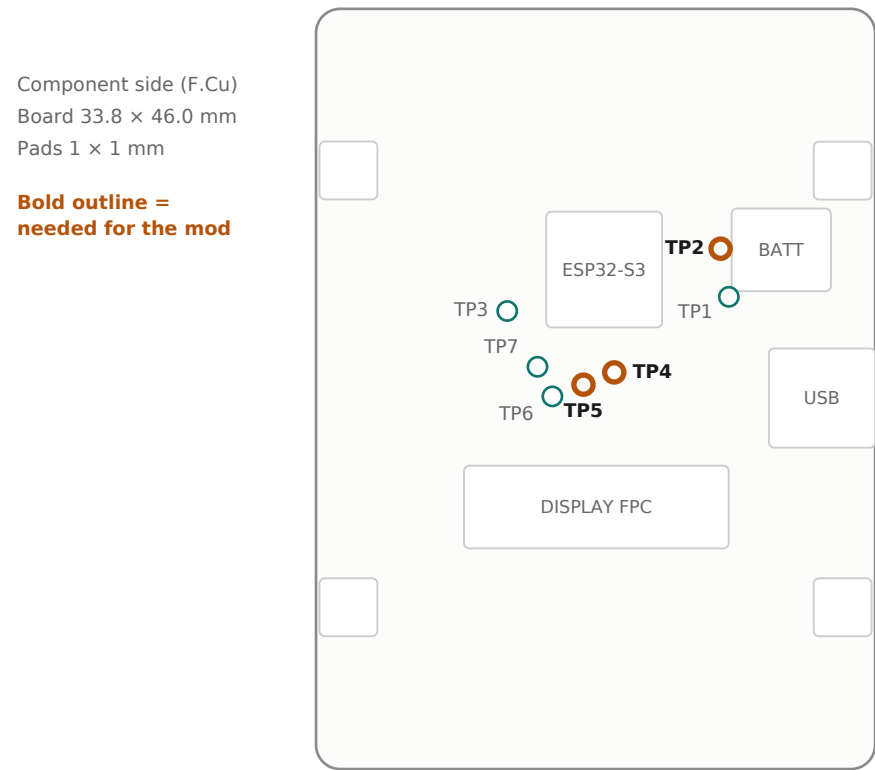
The other four test points are identified along the way, because eliminating them is what makes the last two unambiguous.

2. Before you start

Step	Why
Disconnect the battery	Removes any short-circuit risk while probing. Pull on the connector housing, never the wires.
Black lead in COM , red in INPUT	The 10 A and μ A/mA jacks are not used anywhere in this procedure.
Clip the black lead to the USB shell	Frees one hand. The 1 mm pads need a steady probe.

Power comes from USB throughout, except in phases 1 and 4 where the board must be de-energised.

3. Where the pads are



Positions extracted from *Watchy.kicad_pcb*, branch v3.0. Origin is the top-left corner of the board outline.

TP	Net	Pin	X (mm)	Y (mm)	Needed
TP1	TX	GPIO43	24.96	17.41	—

TP2	RX	GPIO44	24.46	14.50	yes — PWM
TP3	BOOT	GPIO0 / BTN3	11.56	18.27	—
TP4	+3V3	—	18.03	21.98	yes
TP5	GND	—	16.17	22.72	yes
TP6	SCL	GPIO11	14.29	23.43	—
TP7	SDA	GPIO12	13.39	21.64	—

TP4, TP5 and TP6 sit on a shallow diagonal running down and to the left, roughly 2 mm apart. TP7 sits above the left end of that row. TP1 and TP2 are in the gap between the ESP32 and the battery connector, TP2 being the one closer to the connector.

4. Meter settings

Phase	Rotary position	Display should show
1	Ω , then Select until the speaker icon appears	continuity beep
2, 3, 5	V with solid and dashed line (DC volts)	volts, autoranging
4	Ω , Select back to plain resistance	ohms, OL when open

If unsure which Select mode you are in: touch the two probes together. A beep means continuity mode; a reading near zero ohms means resistance mode.

5. Procedure

Phase 1 — find GND (board unpowered, USB out)

Continuity mode. Black probe on the USB shell, red probe on each pad in turn. **A beep means GND.** Exactly one test point should do this — that is TP5. Battery pads will also beep on the negative terminal.

From here on, the identified GND pad is your reference for every voltage measurement.

Phase 2 — voltages (USB in, battery still out)

DC volts. Black on GND, red on each remaining pad.

Reading	Meaning
0.00 V	another GND
3.2 – 3.4 V, rock steady	+3V3 → this is TP4
4.0 – 4.3 V	battery pad — do not solder here
around 3.3 V, less steady	a signal pad — continue with phase 3

With the battery removed, the charger regulates into an open circuit, so battery pads sit near 4.2 V. That makes them easier to tell from 3V3 than they would be with a battery fitted. Confusing the two is the one mistake that destroys the frontlight LEDs.

Phase 3 — eliminate BOOT

Still on DC volts. Rest the red probe on a signal pad and press the **Up** button. If the reading drops to 0 V and returns on release, that pad is **TP3 (BOOT)** — on the v3, GPIO0 is wired to the Up button.

Phase 4 — separate the I²C pads (USB out again)

Resistance mode, board de-energised — a live board gives meaningless resistance readings. One probe on the TP4 pad found in phase 2, the other on each remaining signal pad.

Reading	Meaning
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a few kΩ	SCL or SDA — TP6 and TP7, via their pull-ups
OL / open	TX or RX — TP1 and TP2, your remaining candidates

Two pads should remain after this phase.

Phase 5 — identify TP2

Flash this sketch, then measure in DC volts with black on GND:

```
void setup() { pinMode(43, OUTPUT); pinMode(44, OUTPUT); }
void loop() {
  digitalWrite(43, HIGH); digitalWrite(44, LOW); delay(2000);
  digitalWrite(43, LOW); digitalWrite(44, HIGH); delay(2000);
}
```

Both remaining pads will swing between 0 V and 3.3 V on a two-second cycle, **in antiphase to each other**. The pad reading 3.3 V while the other reads 0 V is **TP2 (GPIO44)** — your PWM connection.

Two pads alternating in exact antiphase occur nowhere else on the board, which is what makes this conclusive rather than suggestive.

Reset the firmware before assembly, otherwise GPIO44 sits at a fixed level and the first frontlight test will mislead you.

6. Record the result

Signal	Test point	Position on the board	Verified
3V3	TP4		
GND	TP5		
PWM	TP2		

Marking the three pads on a photo of your own board pays for itself when you come back to solder.

7. If something does not add up

Symptom	Likely cause
Two pads read a steady 3.3 V and both are open to 3V3	One may be an unpopulated position rather than a test point. Check against the map in section 3 before assuming a wiring difference.
Nothing beeps against the USB shell	Shell may be anodised or coated. Use the negative battery terminal as the ground reference instead.
No pad responds in phase 5	The sketch may not have flashed, or the board is a clone with a different pinout. Confirm with a blink sketch on a known pin first.
Resistance readings drift	The board is still powered. Unplug USB for phase 4.

Companion documents: [watchy-v3-frontlight-mod.md](#) (build guide) and [watchy-v2-frontlight-appendix.md](#) (v2 and USB-C clones).